

# Supporting Information

## Rapid Redox Cycling of Fe(II)/Fe(III) in Microdroplets during Iron-Citric Acid Photochemistry

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## 64 **Text S1. Additional experimental details**

### 65 S1.1 Fe(III)-CA-microdroplet photochemical experiment with catalase

66 It is known that catalase can efficiently decompose  $\text{H}_2\text{O}_2$  to  $\text{H}_2\text{O}$  and  $\text{O}_2$ . To make  
67 clear the role of  $\text{H}_2\text{O}_2$  on the reoxidation of Fe(II) during Fe(III)-CA-microdroplet  
68 photochemical reactions, excess catalase (50  $\mu\text{L}$ ) was added to the 100 mL Fe(III)-CA  
69 solution before spray. The process of microdroplet photoreaction and sample collection  
70 was the same as the experiments without catalase.

### 71 S1.2 Dark reoxidation of photogenerated Fe(II) in Fe(III)-CA-microdroplet samples

72 Samples used for dark reaction were collected from the microdroplet  
73 photoreactions. After 15 min of irradiation, samples were quickly collected from the  
74 bottom of the reactor into a 15 mL centrifuge tube covered with aluminum foil. For the  
75 dark reaction under  $\text{N}_2$  condition, 10 mL aforementioned samples (containing DO) was  
76 then injected to a 15 mL sealed glass bottle which was covered with aluminum foil and  
77 filled with  $\text{N}_2$  in advance. Throughout the dark reaction, 100 sccm  $\text{N}_2$  was kept bubbled  
78 into the glass bottle. For the dark reaction in the aerobic environment, no  $\text{N}_2$  was  
79 introduced into the glass bottle. 2mL of samples was taken out at given time intervals  
80 to quantify the concentration of Fe(II).

### 81 S1.3 pH measurement

82 pH values of both initial solution and samples collected after photoreactions were  
83 measured by a FE28-Standard pH-meter (METTLER TOLEDO, Switzerland) equipped  
84 with a LE438 combination electrode and calibrated with pH 4.01, 7.00, 10.01 standard

85 buffers (METTLER TOLEDO, Switzerland). All solutions were adjusted to pH 2.8 by  
86 1 M HClO<sub>4</sub> before used for photochemical reactions. With the degradation of CA or  
87 BA, pH of all the systems in this study rose slightly and the final pH was near to 3.0,  
88 guaranteeing sediment of iron was absence during reactions.

## 89 **Text S2. HPLC analysis for CA and BA**

90 The concentrations of CA and BA in samples collected from microdroplets and  
91 bulk solutions were all determined by HPLC [Agilent (1200 Series), Dikma Spursil C18  
92 reversed phase column (250 × 4.6 mm, 5 μm)] with the same injection volume (10 μL),  
93 flow rate (0.5 mL/min) and column temperature (25 °C). Differently, for CA, the mobile  
94 phase consisted of 10/90 (v/v) MeOH/NaH<sub>2</sub>PO<sub>4</sub> solution (25 mM) acidified with H<sub>3</sub>PO<sub>4</sub>  
95 to pH 2.5. The detection wavelength was at 214 nm and the retention time of CA was  
96 7.88 min. For BA, the mobile phase consisted of 10/90 (v/v) MeOH/CH<sub>3</sub>COONH<sub>4</sub>  
97 solution (20 mM). The detection wavelength was at 230 nm and the retention time of  
98 BA was 12.30 min. According to the concentration of CA or BA measured by HPLC,  
99 the degradation degree of these two carboxylic acids was calculated by the initial value  
100 before reaction (C<sub>0</sub>) and the residual value after reaction (C).

$$102 \quad \text{Degradation of CA or BA (\%)} = (C_0 - C) / C_0 \times 100\%$$

### **Text S3. Iron concentration measurement by the 1,10-phenanthroline method**

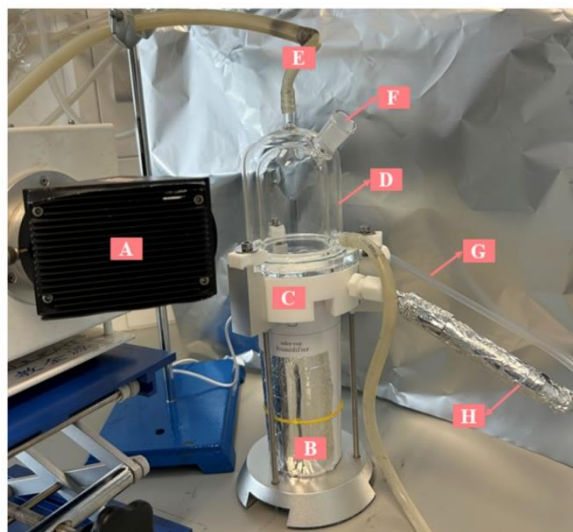
The modified 1,10-phenanthroline method by ammonium fluoride (NH<sub>4</sub>F) was used to shield the coordination effect of Fe(III) during the measurement of Fe(II) concentrations. For this analysis, 0.5 mL of 1,10-phenanthroline solution (C<sub>12</sub>H<sub>8</sub>N<sub>2</sub>, 5 mM), 1 mL of sodium acetate/acetic acid buffer (NaAc/HAc, 1 M, pH 5.0) and 0.5 mL of ammonium fluoride solution (NH<sub>4</sub>F, 0.5 M) were premixed. 1 mL of sample once gathered, was immediately added into the above-mentioned solution. After 15 min of standing in the dark, the mixture solution was put into a 1-cm quartz cell and the absorbance (Abs) at 510 nm was recorded using a Hitachi U-3900 spectrophotometer. For the concentration measurement of the total iron ion (T-Fe), 0.5 mL of hydroxylamine hydrochloride solution (NH<sub>2</sub>OH·HCl, 5%, w/w) was used to replace the ammonium fluoride solution (NH<sub>4</sub>F, 0.5 M), which reduced all iron in solution to the state of Fe(II). The ratio of Fe(II) in both microdroplet- and bulk- samples was calculated by the concentration of Fe(II) and T-Fe. The standard curve of Fe(II) concentration is shown in Figure S5a with ammonium iron(II) sulfate hexahydrate ((NH<sub>4</sub>)<sub>2</sub>Fe(SO<sub>4</sub>)<sub>2</sub>·6H<sub>2</sub>O) as the Fe(II) source. Because the coordination ability of 1,10-phenanthroline with Fe(II) is stronger than CA and BA, the existence of CA or BA can hardly influence the measurement of iron concentration.

$$\text{Ratio of Fe(II) (\%)} = C_{\text{Fe(II)}} / C_{\text{T-Fe}} \times 100\%$$

**Text S4. DO concentration measurement during the photochemical reaction of Fe(III)-CA-bulk solution**

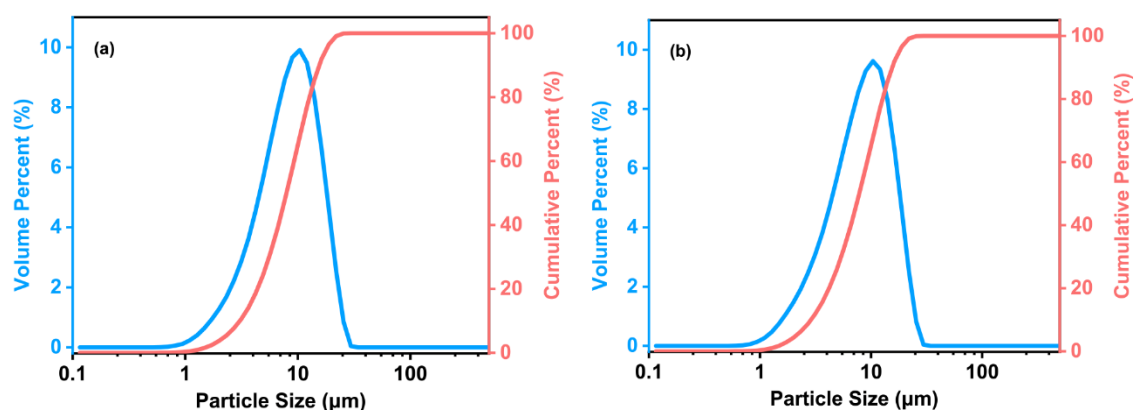
To accurately measure the changes of DO concentration in bulk solution, a custom-made PTFE cap was used to allow the insertion of detection electrode (Thermo Scientific Orion Star A (310D-01A) detector) and gas tube into the Fe(III)-CA aqueous phase. Before irradiation, 100 sccm of dry O<sub>2</sub> was bubbled into bulk solution for more than 30min and DO concentration measurement was started at 10min before irradiation to ensure the state of DO saturation. Then, a xenon lamp with a UV reflector (CEL-HXF 300, Beijing Ceaulight, China) was turned on to initiate the reaction. During Fe(III)-CA-bulk solution photoreaction, 100 sccm of dry O<sub>2</sub> was still kept purged and the DO concentration was recorded at specific time (the same time points as Figure S6) until CA was exhausted. Magnetic stir and circulating water were applied throughout the experiments. To exclude the effect of solvent volatilization, the concentration of total iron was measured both before and after reaction. Figure S15 shows the obvious DO consumption in bulk solution during Fe(III)-CA photochemistry.

## Supporting Figures



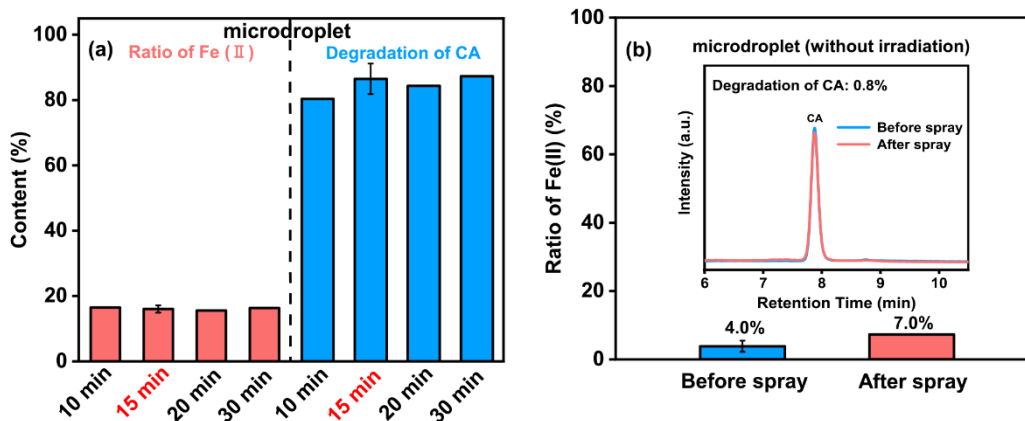
**Figure S1.** Experimental setup for microdroplet photoreaction. Letters in the picture represent different components of microdroplet photoreaction device. Concretely, A-xenon lamp equipped with a UV reflector film, B-ultrasonic vibration sprayer, C-droplet storage trough, D-double-walled quartz photoreactor, E-circulating cooling water, F-gas outlet, G-gas inlet, H-product collection pipe. The total pressure inside the photochemical reactor was kept at 1 atm throughout all the microdroplet experiments.



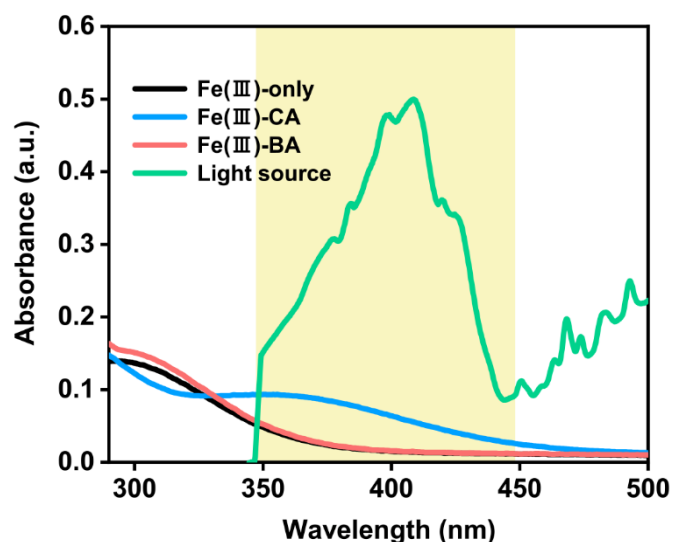


**Figure S2.** The size distribution of microdroplets produced from the ultrasonic sprayer.

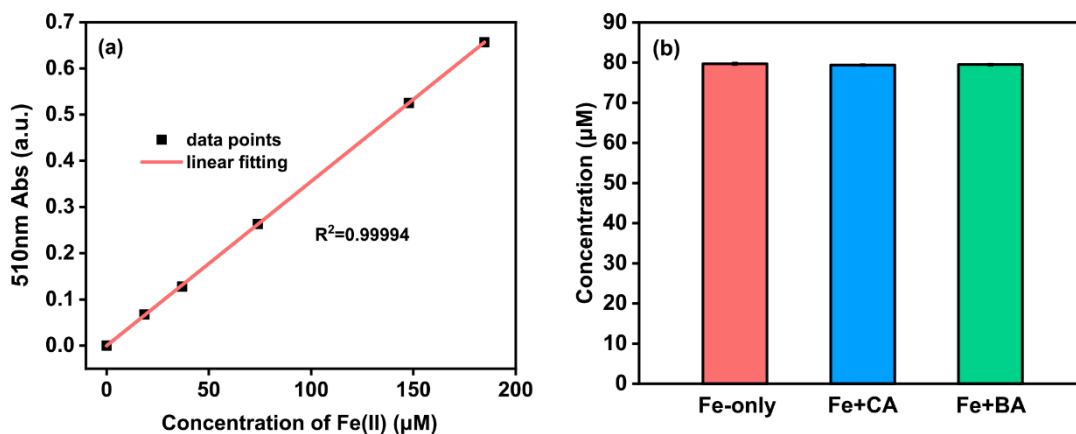
A laser particle size analyzer (PW 180-C, Beijing Haixin Rui Technology, China) was used to measure the microdroplet size at 7 cm above the sprayer (namely, at the middle of the microdroplet reactor). (a)  $[\text{Fe(III)}]_0 = 80 \mu\text{M}$ ,  $[\text{CA}]_0 = 500 \mu\text{M}$ . (b)  $[\text{Fe(III)}]_0 = 80 \mu\text{M}$ ,  $[\text{BA}]_0 = 80 \mu\text{M}$ .  $\text{pH} = 2.8 \pm 0.2$ .



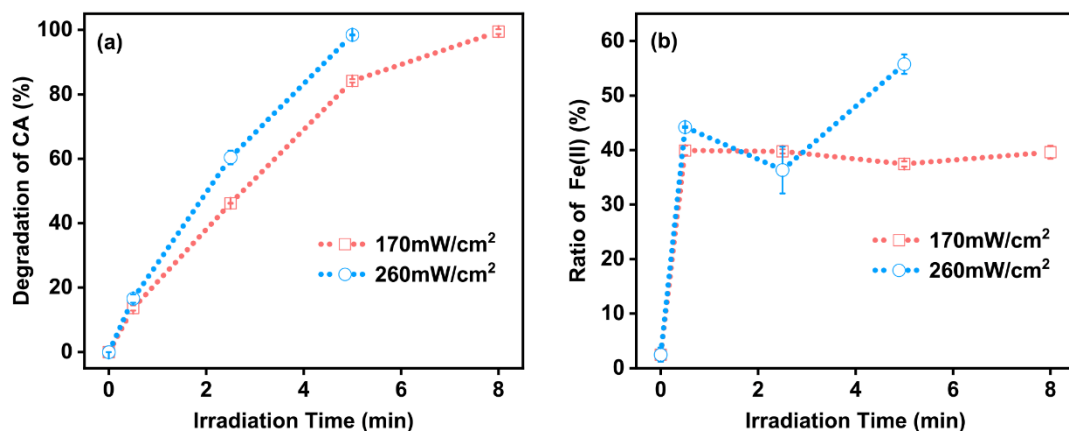
**Figure S3.** Performance verification of microdroplet photochemical device. (a) The effect of irradiation time on Fe(II) ratio and CA degradation in Fe(III)-CA-microdroplet system at  $I = 260 \text{ mW/cm}^2$ . (b) Dark atomization experiment of Fe(III)-CA solution. Besides the process in the photoreactor conducted without irradiation, other procedures like spray and collection in the dark experiment were the same as the microdroplet photoreaction.  $[\text{Fe(III)}]_0 = 80 \text{ }\mu\text{M}$ ,  $[\text{CA}]_0 = 500 \text{ }\mu\text{M}$ ,  $\text{pH} = 2.8 \pm 0.2$ , pure oxygen atmosphere. Total flow rate of gas was kept at 100 sccm, and the total pressure in microdroplet reactor was 1 atm. It should be noted that 4% of Fe(II) is found in the solution before spray. This Fe(II) may come from the impurity of the used reagent, which is Reagent Grade. To guarantee the repeatability of the experiments and the uniformity of the initial solution, we always measured the concentration of total iron and Fe(II) in newly prepared solution by the 1,10-phenanthroline method before spray. The Fe(II) ratio remained constant in every measurement. Therefore, the error caused by this initial concentration of Fe(II) is a systematic one. In addition, after spray and collection in dark, there is a slight increase of Fe(II) ratio (about 3%). Such a tiny increase in Fe(II) ratio should be attributed to the reaction caused by the ultrasonic spray, which is known to produce some active species to initiate weak reaction.



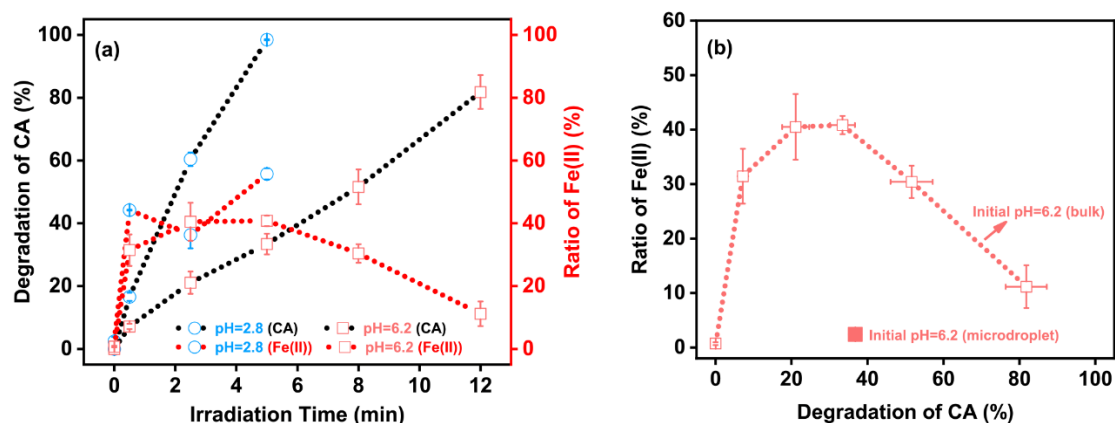
**Figure S4.** Emission spectrum of the xenon lamp equipped with a UV reflector film and UV-vis absorption spectra of three iron containing solution. Emission spectrum was measured by an USB-4000-VIS-NIR Ocean Optics spectrometers. To present in the same figure, the intensity of light source emission spectrum was normalized by the maximum value and divided by a factor of 2.  $[\text{Fe(III)}]_0 = 80 \mu\text{M}$ ,  $[\text{CA}]_0 = 500 \mu\text{M}$ ,  $[\text{BA}]_0 = 80 \mu\text{M}$ . All solutions were adjusted to pH 2.8 by 1 M  $\text{HClO}_4$ . CA formed complexes with Fe(III), making the absorption bands move to the range of 350-450 nm but the presence of BA in solution almost did not change the UV-vis spectra of Fe(III)-only solution, which indicated a fairly weak coordination between BA and Fe(III).



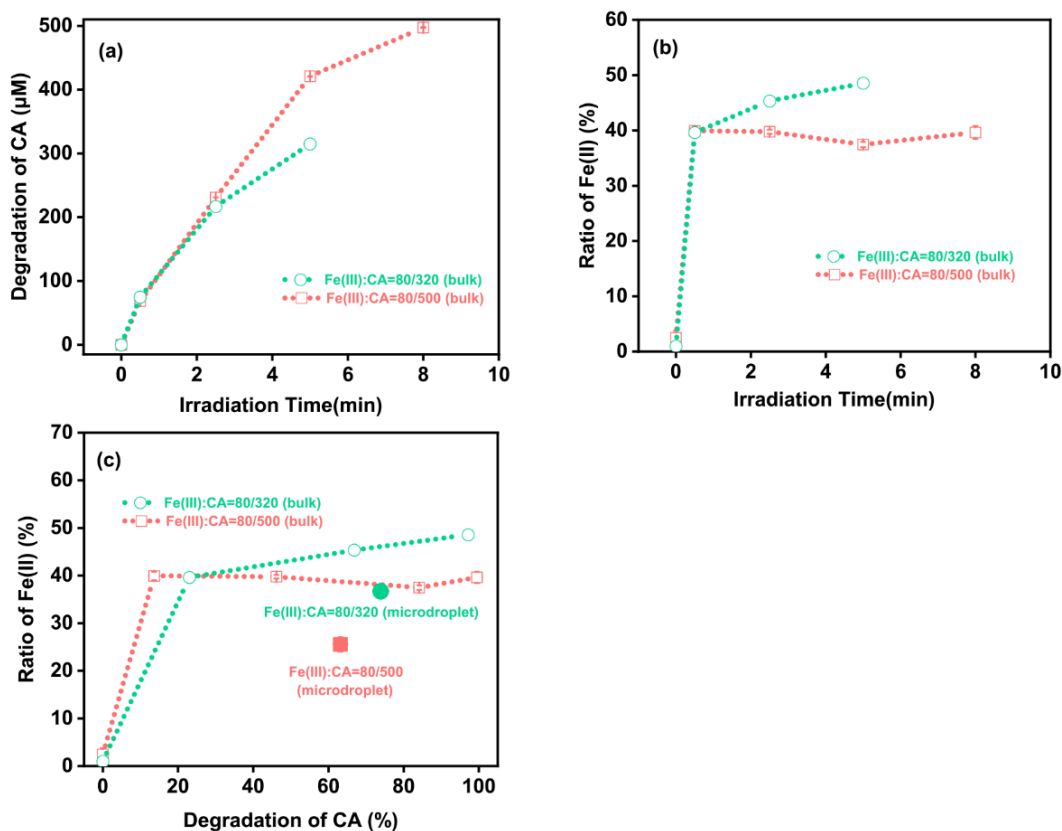
**Figure S5.** Iron ion concentration measurement by the 1,10-phenanthroline method. (a) The calibration curve of Fe(II) concentration.  $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$  was used as the ferrous source. (b) Measured concentration of Fe with and without CA or BA. In the Fe-only experiment, the solution of 80  $\mu\text{M}$   $\text{Fe}(\text{ClO}_4)_3$  was directly measured. In the Fe + CA or Fe + BA experiment, 500  $\mu\text{M}$  CA or 80  $\mu\text{M}$  BA was added into 80  $\mu\text{M}$   $\text{Fe}(\text{ClO}_4)_3$ . Newly prepared above solutions were immediately used to Fe concentration determination by 1,10-phenanthroline method and repeated at least twice.



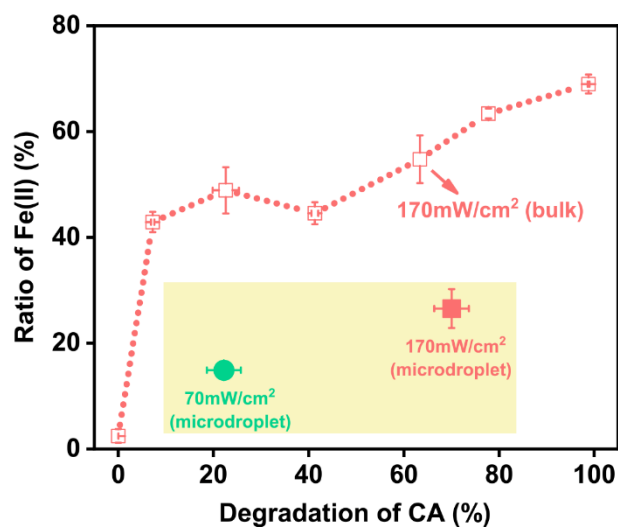
**Figure S6.** CA degradation (a) and Fe(II) ratio change (b) during Fe(III)-CA-bulk photochemistry at two different light intensities.  $[\text{Fe(III)}]_0 = 80 \mu\text{M}$ ,  $[\text{CA}]_0 = 500 \mu\text{M}$ ,  $\text{pH} = 2.8 \pm 0.2$ , 100 sccm oxygen bubbling. The total pressure in the reactor was 1 atm.



**Figure S7.** Photochemistry of Fe(III)-CA system at pH close to neutral. (a) CA degradation and Fe(II) concentration change. (b) The comparison between Fe(III)-CA-microdroplet system (solid point) and Fe(III)-CA-bulk phase (hollow points) under pure O<sub>2</sub>. Total flow rate of gas was kept at 100 sccm, and the total pressure in each reactor was 1 atm. [Fe(III)]<sub>0</sub> = 80 μM, [CA]<sub>0</sub> = 500 μM, I = 260 mW/cm<sup>2</sup>, pH = 6.2 ± 0.1 (adjusted by 1 M NaOH).

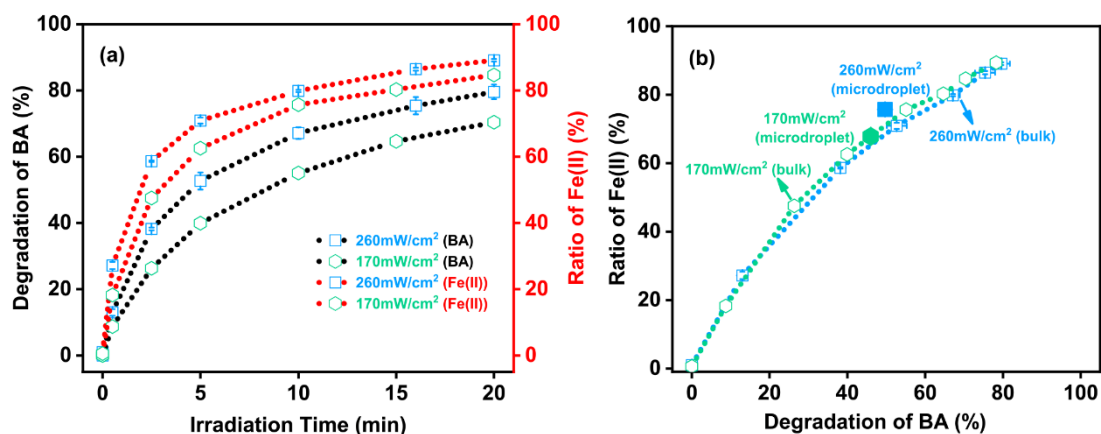


**Figure S8.** The effect of initial concentration ratio of Fe(III) to CA in Fe(III)-CA photochemistry. CA degradation (a) and Fe(II) ratio change (b) in bulk solution. (c) The comparison between Fe(III)-CA-microdroplet system (solid points) and Fe(III)-CA-bulk phase (hollow points). Total flow rate of  $\text{O}_2$  was kept at 100 sccm, and the total pressure in each reactor was 1 atm. The initial concentrations ( $\mu\text{M}$ ) of Fe(III) and CA have already been marked in the figure directly.  $I = 170 \text{ mW}/\text{cm}^2$ ,  $\text{pH} = 2.8 \pm 0.2$ .

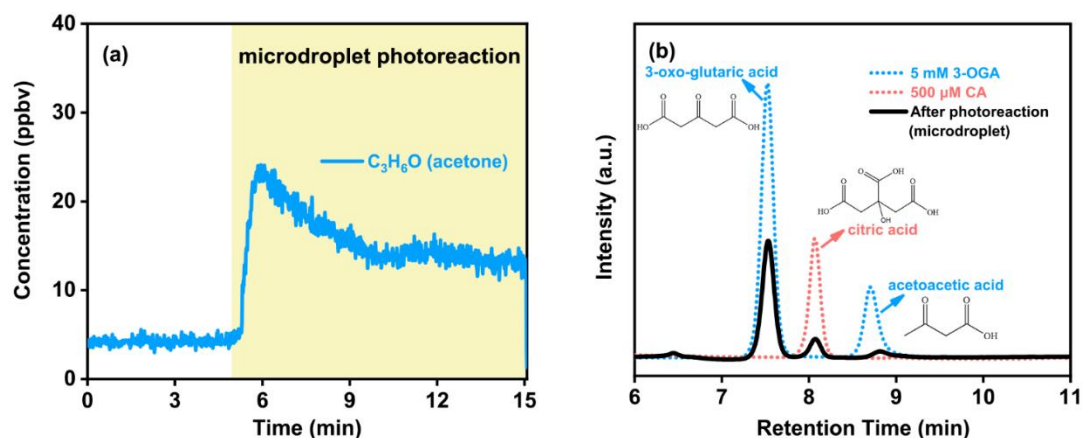


**Figure S9.** The comparison between Fe(III)-CA-microdroplet system (solid points) and Fe(III)-CA-bulk phase (hollow points) at a lower initial concentration ratio of Fe(III) to CA under O<sub>2</sub> (100%) atmosphere. Total flow rate of O<sub>2</sub> was kept at 100 sccm, and the total pressure in each reactor was 1 atm. [Fe(III)]<sub>0</sub> = 40 μM, [CA]<sub>0</sub> = 500 μM, pH = 2.8 ± 0.2.

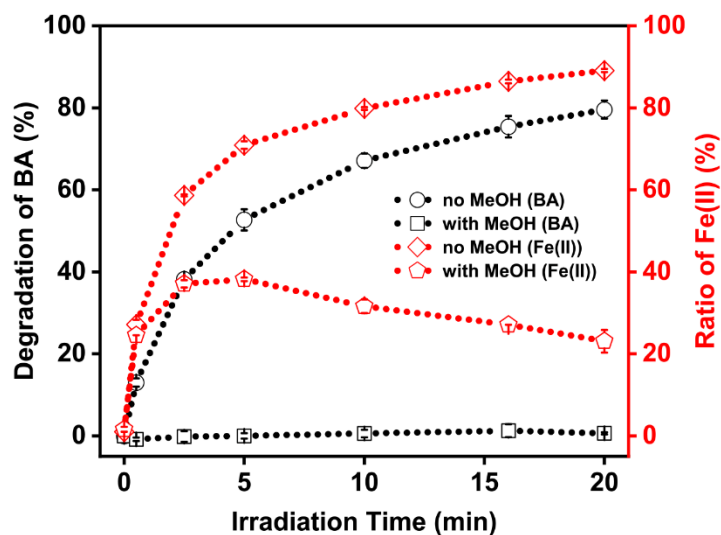




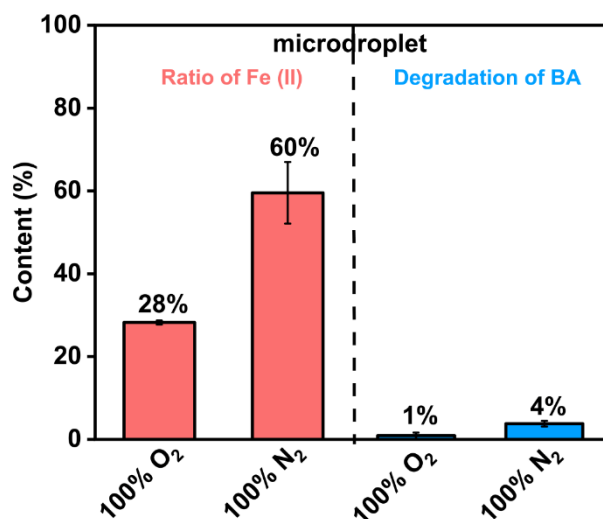
**Figure S10.** The effect of light intensity on Fe(III)-BA-bulk photochemistry. (a) BA degradation and Fe(II) ratio change at two different light intensities. (b) The comparison between microdroplet system (solid points) and bulk phase (hollow points).  $[\text{Fe(III)}]_0 = 80 \mu\text{M}$ ,  $[\text{BA}]_0 = 80 \mu\text{M}$ ,  $\text{pH} = 2.8 \pm 0.2$ , 100 sccm oxygen bubbling. The total pressure in each reactor was 1 atm.



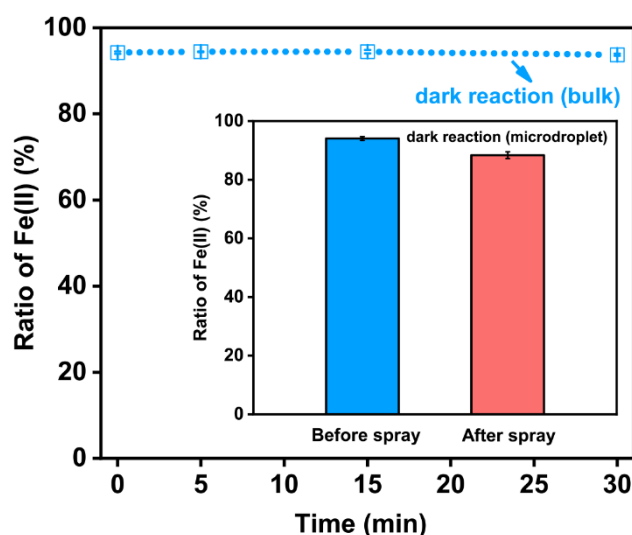
**Figure S11.** The detection on the photodegradation products of CA in Fe(III)-CA-microdroplet photochemical system. (a) The production and release of gaseous acetone. The PTR-ToF-MS spectra was recorded at a time resolution of 1 s with inlet flow rate of 50 sccm.  $\text{H}_3\text{O}^+$  ionization mode was used with the drift voltage of 600 V, drift temperature of 70 °C, and drift pressure of 2.20 mbar. (b) HPLC analysis for the collected liquid samples and the standard samples of 3-oxo-glutaric acid and CA.  $[\text{Fe(III)}]_0 = 80 \text{ } \mu\text{M}$ ,  $[\text{CA}]_0 = 500 \text{ } \mu\text{M}$ ,  $\text{pH} = 2.8 \pm 0.2$ ,  $I = 260 \text{ mW/cm}^2$ , pure  $\text{O}_2$  atmosphere. Total flow rate of gas was kept at 100 sccm, and the total pressure in the reactor was 1 atm.



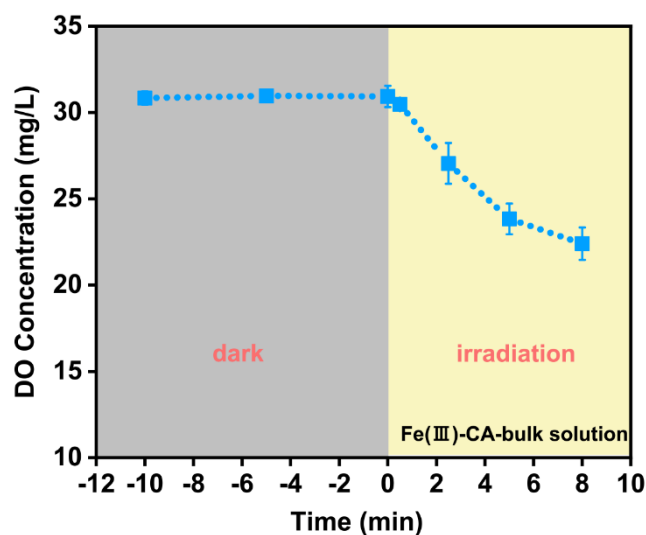
**Figure S12.** The effect of MeOH on BA degradation rate and Fe(II) ratio in Fe(III)-BA-bulk photochemical system.  $[\text{Fe(III)}]_0 = 80 \mu\text{M}$ ,  $[\text{BA}]_0 = 80 \mu\text{M}$ ,  $[\text{MeOH}]_0 = 0.25 \text{ M}$ ,  $\text{pH} = 2.8 \pm 0.2$ ,  $I = 260 \text{ mW/cm}^2$ , 100 sccm  $\text{O}_2$  bubbling. The total pressure in the reactor was 1 atm.



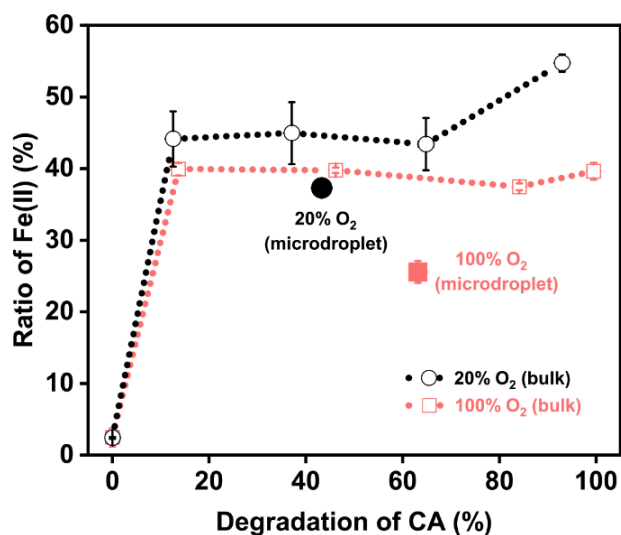
**Figure S13.** The effect of O<sub>2</sub> partial pressure on Fe(II) ratio and BA degradation in Fe(III)-BA-MeOH microdroplet system. [Fe(III)]<sub>0</sub> = 80 μM, [BA]<sub>0</sub> = 80 μM, [MeOH]<sub>0</sub> = 0.25 M, pH = 2.8 ± 0.2, I = 260 mW/cm<sup>2</sup>. Total flow rate of gas was kept at 100 sccm, and the total pressure in the reactor was 1 atm.



**Figure S14.** The oxidation of Fe (II) by molecular oxygen under dark condition in Fe(II)-CA bulk solution. Inset: The oxidation of Fe (II) by molecular oxygen under dark condition in Fe(II)-CA microdroplet system. Total flow rate of O<sub>2</sub> was kept at 100 sccm, and the total pressure in each reactor was 1 atm. [Fe(II)]<sub>0</sub> = 80 μM, [T-Fe] = 85 μM, [CA]<sub>0</sub> = 500 μM, pH = 2.8 ± 0.2. The about 5% of Fe(III) in Fe(II)-CA bulk solution should come from the impurity of the used reagent, which is Reagent Grade (Fe(ClO<sub>4</sub>)<sub>2</sub>·xH<sub>2</sub>O, Aladdin). The concentrations of Fe(II) remain stable during the dark reaction, and exhibit good repeatability among different experimental run. Therefore, the error caused by this initial concentration of Fe(II) is a systematic one. The slight decrease in the ratio of Fe(II) after spray should be attributed to the reaction caused by the active species during ultrasonic spray.



**Figure S15.** DO concentration change in Fe(III)-CA bulk solution during bubbling O<sub>2</sub> in the dark and photoreaction. [Fe(III)]<sub>0</sub> = 80 μM, [CA]<sub>0</sub> = 500 μM, pH = 2.8 ± 0.2, I = 170 mW/cm<sup>2</sup>, 100 sccm oxygen bubbling. The total pressure in the reactor was 1 atm. At 8 min of irradiation, CA was nearly depleted (see Figure S6a), and the record of DO concentration was stopped.



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360 **Figure S16.** The comparison between Fe(III)-CA-microdroplet system (solid points)

361 and Fe(III)-CA-bulk phase (hollow points) under different O<sub>2</sub> partial pressure. Total

362 flow rate of gas was kept at 100 sccm, and the total pressure in the reactor was 1 atm.

363 [Fe(III)]<sub>0</sub> = 80 μM, [CA]<sub>0</sub> = 500 μM, I = 170 mW/cm<sup>2</sup>, pH = 2.8 ± 0.2.

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365 **References**

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